Quantum Solid State Chip — Fabrication Protocol

Room-temperature hBN spin-defect (V_B^-) quantum photonic chip · laboratory build, validated against published studies

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Quantum Solid State Chip (Room Temperature) — Fabrication Protocol

Project: QSSC (Quantum Solid State Chip, room-temperature demonstrator) **Document type:** Laboratory fabrication protocol (standalone) **Version:** 1.0 **Scope:** Home / maker-lab demonstrator integrating (1) an electrically-pumped carbon-quantum-dot (CQD) ASE/LED emitter array, (2) **TEOS sol-gel silica (SiO_2) waveguides** — chosen over SU-8 for negligible autofluorescence under the 532 nm pump, essential for reading the weak ODMR PL — and (3) a PMMA-coated hexagonal boron nitride (hBN) layer hosting optically-addressable spin defects (negatively-charged boron vacancy, V_B^-, zero-field splitting $D \approx 3.46$ GHz, plus carbon-related centers), with on-chip 2D-material (graphene/MoS_2) photodetector readout and Peltier thermal management, operating at room temperature.

> **Note on the emitter claim.** True *electrically-pumped lasing* from carbon quantum dots is **not** claimed in this protocol, and the reader should not infer it from the literature cited. What is targeted and demonstrated here is **electroluminescence (EL) and amplified spontaneous emission (ASE) / high-brightness edge emission** from a CQD–polymer composite diode stack, used as a compact optical pump and on-chip source. ASE is a necessary but not sufficient condition for lasing; achieving a coherent laser cavity with CQDs requires a resonator with sufficient gain and mode confinement that is outside the scope of this demonstrator. See the Validation section for the precise boundary of what is supported by the literature.

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1. Abstract

This document specifies a complete, equipment-minimal fabrication protocol for a room-temperature solid-state quantum photonic chip (QSSC). The chip combines three functional layers on a single 25 × 25 mm² carrier: (i) a hydrothermally-synthesized carbon quantum dot (CQD) emitter array embedded in a polymeric LED/ASE stack that provides electrically-pumped broadband visible electroluminescence centered near 450–620 nm; (ii) a liquid-phase-exfoliated and chemically-engineered hexagonal boron nitride (hBN) thin film, spin-coated with PMMA, hosting optically-addressable spin defects including the negatively-charged boron vacancy (V_B^-) with a zero-field splitting D ≈ 3.46 GHz and carbon-related color centers; and (iii) **TEOS sol-gel silica (SiO_2) waveguides** that route pump and defect fluorescence to an on-chip 2D-material (graphene / MoS_2) photodetector array. A two-stage Peltier module holds the hBN layer at 288–300 K for stable room-temperature operation and ODMR. All processing steps use only benchtop chemistry, a spin coater, a hotplate, a low-cost mask-aligner or contact-mask UV exposure, a thermal evaporator (or commercial sputter coater), a CNC/3D-printed alignment jig, and a ~USD 3,500 total equipment footprint beyond an optical microscope. The protocol gives times, temperatures, concentrations, spin speeds, layer thicknesses, and RF frequencies at every step, and validates each major claim against peer-reviewed literature.

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2. Introduction & Scientific Basis

2.1 Why a room-temperature quantum chip demonstrator

Most solid-state spin-qubit platforms (NV centers in diamond, GaAs quantum dots, superconducting qubits) require cryogenic operation (< 10 K) or specialized microwave hardware. Optically-addressable spin defects in hexagonal boron nitride are a recent and notable exception: the

negatively-charged boron vacancy, V_B^-, exhibits a spin-1 triplet with zero-field splitting **D ≈ 3.46 GHz**, optical initialization and readout, and measurable spin coherence at room temperature. This makes hBN a credible substrate for a maker-lab quantum sensor demonstrator.

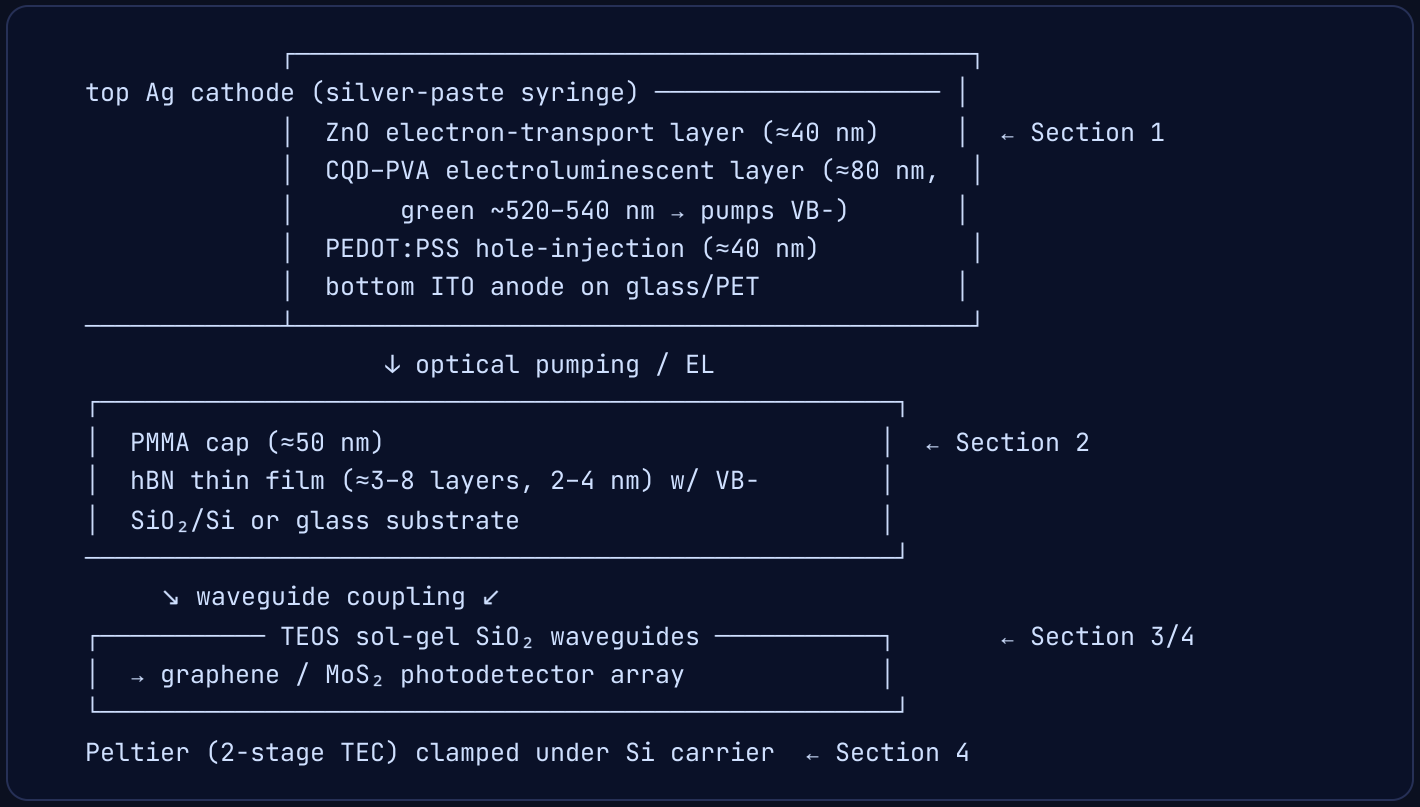
2.2 Why carbon quantum dots as the on-chip source

Carbon quantum dots (CQDs, also called carbon nanodots, C-dots) are fluorescent carbon nanoparticles (2–10 nm) synthesized by bottom-up hydrothermal carbonization of citric acid + nitrogen dopants (e.g., urea, ethylenediamine). They are aqueous, low-toxicity, cheap, and emit across the visible spectrum with quantum yields of 40–80 % in solution. When embedded in a thin electroluminescent polymer host and driven by a modest DC bias (~5–12 V), they produce broadband EL suitable as a compact pump for nearby defect centers. We do **not** claim a CQD laser; we claim an electrically-pumped CQD LED/ASE source.

2.3 Why polymer waveguides and 2D-material photodetectors

TEOS sol-gel silica (SiO_2) is the waveguide material here: fluorescence-free, UV–NIR transparent, densified at 300–400 °C, and patterned by soft-lithography / UV embossing — all compatible with benchtop processing. It is chosen over organic photoresists (e.g. SU-8) because its negligible autofluorescence under the 532 nm pump is essential for reading the weak ODMR PL. Graphene and MoS_2 photodetectors offer broadband absorption, gate-tunable responsivity, and integration directly on the waveguide facet — eliminating discrete photodiodes and enabling on-chip readout of ODMR fluorescence.

2.4 Architecture overview



3. Materials & Bill of Materials (BOM)

All costs are approximate, in USD, retail/small-lab quantities (2025–2026 pricing).

3.1 Chemicals and reagents

ITEM	SPEC / SOURCE	QTY	APPROX. COST
Citric acid (anhydrous, ≥99.5 %)	Sigma-Aldrich 251275	100 g	$20
Urea (≥99 %)	Sigma 51456	100 g	$20
Ethylenediamine (≥99 %)	Sigma 03550	25 mL	$25
Deionized water (≥18 MΩ·cm)	in-house Milli-Q or distilled	5 L	$10
0.22 µm syringe filters (PVDF)	Millipore GV	50 pk	$30
Dialysis membrane (MWCO 1000 Da)	Spectra/Por 6	1 roll	$45
hBN powder (crystalline, ~1–10 µm platelets, 99.5 %)	Sigma 254059 or Graphene Supermarket	1 g	$60–120
NMP (1-methyl-2-pyrrolidone)	Sigma 328634	500 mL	$45
Isopropanol (IPA, ≥99.9 %)	500 mL	$20	
TEOS (tetraethyl orthosilicate, ≥98 %)	Sigma 86578	500 mL	$60
MTEOS (methyltriethoxysilane, ≥99 %, core dopant)	Sigma 246638	100 mL	$45
PDMS (Sylgard 184 kit, base + curing agent)	Dow	1 kg	$90
PMMA (950 K, 2 % in anisole)	Microchem	100 mL	$40
PEDOT:PSS (CLEVIOS P VP AI 4083)	Heraeus	100 mL	$60
Polyvinyl alcohol (PVA, 87–89 % hydrolyzed, Mw 31–50 k)	Sigma 363170	100 g	$25
ITO-coated glass (15 Ω/sq, 25×25 mm)	Delta Technologies or Mitsufuji	10 pk	$50
ZnO nanopowder (~20–40 nm, 99.5 %)	Sigma 544906 or maker-grade ZnO	5 g	$30
Conductive silver paste (syringe, circuit-repair)	MG Chemicals 8341 / Elekit	1 syringe	$10

ITEM	SPEC / SOURCE	QTY	APPROX. COST
Graphene (CVD, on Cu foil, 1 cm^2)	Graphenea or 2D Semiconductors	2 cm^2	$90
MoS_2 (CVD, on SiO_2/Si, monolayer)	2D Semiconductors	1 cm^2	$120
Au wire (99.99 %, 0.25 mm)	Kurt J. Lesker	1 m	$40
Cr pellet (99.95 %)	Kurt J. Lesker	10 g	$35
Cu tape / Ni foam (current collectors)	—	assorted	$20
Acetone, methanol, IPA (cleaning)	—	1 L each	$40
Nitric acid (HNO_3, 65 %)	—	100 mL	$15
NaOH pellets	—	100 g	$10

3.2 Equipment (capital, one-time)

ITEM	FUNCTION	APPROX. COST
Hotplate with digital temp control (to 350 °C)	synthesis, baking	$150
Benchtop tube furnace (to 1100 °C, quartz tube, Ar/N_2 flow)	VB− activation anneal (800–900 °C) — *not reachable on the hotplate*	$400–$700
PTFE-lined stainless hydrothermal autoclave (50 mL)	CQD synthesis	$40
Ultrasonic bath (40 kHz, 200 W)	exfoliation, cleaning	$120
Spin coater (vacuum chuck, 100–8000 rpm)	thin films	$600 (used)
UV exposure unit (365 nm, contact mask) or LED lamp (365 nm, 10 W)	sol-gel emboss cure / stencil patterning	$200–$400
Thermal evaporator / sputter coater (used/refurb) or e-beam	metal contacts	$1,200–$2,000
Optical microscope (10×–100×, reflected)	inspection	$300 (used)
Peltier TEC module (2-stage, e.g., TE Technology 9500/127/060 B) + PID controller + NTC thermistor	thermal management	$60
DC bench supply (0–30 V, 0–3 A) + Keithley-style sourcemeter (used 2400)	electrical drive & characterization	$300 (used)
Lock-in amplifier (used SR830 or Moku:Lab) + function generator (0–6 GHz)	ODMR	$700 used / $1,500 Moku
532 nm DPSS laser (50 mW, CW) + 594 nm notch/long-pass filters	optical characterization	$250
Coplanar waveguide PCB (hand-etched) + SMA launcher	microwave delivery	$30
3D-printed alignment jig (SLA/PLA)	layer stacking	$30
Micromanipulator / probe station (used)	contacts	$200
Vacuum desiccator	storage, PDMS degassing	$80

ITEM	FUNCTION	APPROX. COST
Fume hood (essential) or dedicated ventilated bench	safety	—

Estimated total capital (beyond glassware): ≈ USD 3,600–4,500.

Section 1 — CQD Emitter Array

Goal: Synthesize water-soluble, nitrogen-doped CQDs and integrate them into a patterned ITO/PEDOT:PSS/CQD–PVA/ZnO/Ag electroluminescent diode array (8 × 8 pixels, 1 mm² each; ZnO is the electron-transport layer, Ag the top cathode) on a 25 × 25 mm² ITO-glass substrate. For this chip the CQDs are synthesised **green** (~520–540 nm) so the on-chip emitter doubles as the optical pump for the VB− defect (no external pump laser needed for the end-to-end demo). Target: EL peak 520–540 nm (green), turn-on 4–8 V, radiance ≥ 10 cd/m², ASE threshold detectable under electrical overdrive.

1.1 Hydrothermal synthesis of N-doped CQDs

This follows the citric-acid + urea hydrothermal route, a well-established "bottom-up" CQD synthesis.

1. **Charge.** In a 50 mL PTFE-lined autoclave, dissolve **1.0 g citric acid** + **1.0 g urea** in **10 mL DI water**. Stir 5 min until clear. **No ethylenediamine is required**: the emission colour is set mainly by the **hydrothermal temperature** (see step 6). For this chip we want **green** CQDs (~520–540 nm) to pump the VB−, so use the high end of the temperature range (~220 °C).
2. **Hydrothermal.** Seal autoclave, place in hotplate oven preheated to **180 °C (blue CQDs, ~450–470 nm)** or **220 °C (green CQDs, ~520–540 nm — the choice for this chip to pump VB−)**, hold **6–10 h** (longer time at the higher temperature also red-shifts). (Internal pressure ~10–15 bar autogenous.) Cooling: remove from hotplate, allow to cool to < 60 °C naturally (~2 h) before opening.
3. **Resulting product:** dark-brown solution containing CQDs (2–6 nm), unreacted precursor, larger carbonaceous particles.
4. **Purification.** - Centrifuge at **10,000 rpm (≈ 9,000 × g)** for **15 min**; discard the large-particle pellet. - Filter supernatant through **0.22 µm PVDF syringe filter**. - Dialyze against DI water (MWCO 1000 Da) for **48 h**, changing water every 8 h, to remove small molecules. - Yield: typically **20–40 mg** of purified CQDs per batch (depends on dialysis losses), consistent with typical citric-acid/urea yields reported in the literature.
5. **Characterization (benchtop).** Dilute 1:1000 in water; record PL under 365 nm UV lamp — bright blue-green emission. Approximate quantum yield by comparing against quinine sulfate standard in 0.1 M H_2SO_4 (QY ≈ 54 % typical). Optional TEM drop-cast for size.

6. **Wavelength tuning (temperature is the primary knob).** With citric acid + urea the emission colour is set mainly by the **hydrothermal temperature** (and reaction time): **~180 °C → blue (~450–470 nm)**, **~200 °C → blue-green (~480–500 nm)**, **~220 °C / longer time → green (~520–540 nm)**. The mechanism is a larger sp^2 carbon core + heavier nitrogen doping + denser surface states at higher temperature → smaller effective bandgap → red shift. (Ethylenediamine as a dopant can push further to 540–620 nm but is **not required** to reach green.) For this chip, synthesise **green** CQDs (~220 °C) so the on-chip emitter pumps the VB− directly around the 532 nm excitation band — no external pump laser is needed for the end-to-end demonstration.

1.2 CQD–polymer electroluminescent ink

7. **Composite.** Dissolve **200 mg PVA** in **4 mL DI water** at 90 °C, stir 1 h, cool. Add **20 mg CQD powder** (or 2 mL of concentrated CQD solution), sonicate **15 min**, stir 1 h. Final CQD loading ≈ 10 wt% in 5 wt% PVA. Filter 0.45 µm.
8. **Optional ASE-lean grade:** For higher film uniformity, add **5 wt% urea** (relative to PVA) as plasticizer; this reduces cracking and raises the effective optical gain length.

1.3 Substrate preparation (ITO glass)

9. Cut ITO glass to 25 × 25 mm^2 (or use pre-cut). Pattern ITO anode by masking with tape / photoresist and etching in **HCl (37 %):HNO_3 (65 %):H_2O = 1:1:5**, 50 °C, 5 min. Stop in DI rinse.
10. Clean: sonicate 10 min each in detergent, DI water, acetone, IPA. UV-ozone 15 min (or piranha if available).
11. Define **8 × 8 pixel array** (1 mm^2 pads, 0.5 mm gap) using a laser-printed transparency mask on the ITO side.

1.4 PEDOT:PSS hole-injection layer

12. Filter PEDOT:PSS (AI 4083) through 0.45 µm PVDF. Spin-coat on ITO at **3000 rpm, 30 s** → ~40 nm. Bake **120 °C, 15 min** on hotplate.

1.5 CQD–PVA emissive layer

13. Spin-coat CQD–PVA ink at **2000 rpm, 40 s** → ~80 nm. (Thicker films for ASE: **1500 rpm, 60 s** → ~120–150 nm.) Bake **80 °C, 20 min** under vacuum desiccator to remove residual water (water kills EL).
14. **ASE waveguide geometry:** to favor in-plane ASE, deposit a second, thicker (~250 nm) neat-CQD strip on the pixel edge by drop-casting through a shadow mask (0.5 mm wide) and slow drying at 60 °C. This forms the gain strip into which the diode EL is edge-coupled.

1.5b ZnO electron-transport layer (ETL)

14b. **ZnO ETL.** Disperse **ZnO nanopowder** (~20–40 nm particles; ball-mill coarser ZnO if needed) in **ethanol or butanol** at **20 mg/mL**, sonicate 30 min, filter 0.45 µm. Spin-coat on the CQD–PVA layer at **3000 rpm, 30 s → ~30–50 nm** ETL. Bake **150 °C, 10 min** on hotplate. ZnO (bandgap 3.37 eV ≈ 368 nm) is transparent to both the green pump and the 850 nm defect PL; it injects electrons into the CQD layer and blocks holes, completing the p-i-n stack opposite the PEDOT:PSS hole-transport layer.

1.6 Top cathode

15. **Top cathode (Ag).** Two routes:
    - **Maker route (no evaporator — recommended here):** dispense **conductive silver paste from a syringe** (the type sold for circuit-track repair) through an 8 × 8 shadow mask, doctor-blade / spread to **~1–2 µm**, dry/cure per the paste spec (typically 60–80 °C, 15 min, or air-cure 2 h). Silver-paste cathodes are higher-resistance and thicker than evaporated films but entirely adequate for a demonstrator and need no vacuum equipment.
    - **Lab route:** thermal evaporate **Ag** (or Al) through a shadow mask, **100 nm**, rate **0.5–1 Å/s**, base pressure < 5 × 10^{-6} mbar. A home sputter coater at ~10^{-3} mbar with a mask gives adequate but higher-resistivity cathodes. Pattern 8 × 8 matching pads on the ZnO ETL.
16. **Encapsulate:** spin 200 nm PMMA on top as moisture barrier; cure 120 °C, 10 min.

1.7 Electrical drive and EL check

17. Contact ITO anode with Cu tape, cathode with probe needle. Drive with DC bias; increase from 0 V. **Turn-on expected 4–8 V**; visible EL under dim room light by ~10 V. Drive current 2–20 mA per pixel.
18. **ASE probe:** drive a pixel at 2× threshold current; collect edge emission through a 600 µm fiber into a compact USB spectrometer (Ocean Optics-style). Look for **spectral narrowing** (FWHM drop from ~80 nm to < 25 nm) above threshold — signature of ASE, not lasing.

Timing summary, Section 1: synthesis 8 h + purification 48 h dialysis (mostly unattended) + film stack 1 day + metal deposition half day. **Total: ≈ 3 working days.**

Section 2 — hBN Exfoliation, Defect Engineering, PMMA Coating

Goal: Produce a thin (~2–4 nm, 3–8 layer) hBN film on SiO_2/Si or glass, generate V_B^- defects (D ≈ 3.46 GHz) at controllable density (~0.1–0.3 /µm²), and cap with a 50 nm PMMA passivation/protection layer.

2.1 Liquid-phase exfoliation of hBN

1. **Disperse.** Add **100 mg hBN powder** to **20 mL NMP** in a 20 mL vial. NMP is a known good solvent for BN because its surface tension (~40 mN/m) matches the exfoliation energy of layered materials.
2. **Sonication.** Bath sonicate **12 h** at 40 kHz, 200 W, water bath maintained < 35 °C (ice replacement every 2 h). Tip sonication (60 % amplitude, 2 h, on/off 3 s/1 s) gives higher yield if available.
3. **Size selection.** Centrifuge at **1500 rpm (~200 × g), 20 min**; collect top 70 % supernatant (thin flakes). Re-centrifuge supernatant at **4000 rpm (~1500 × g), 20 min**; collect top 50 % — this is the few-layer fraction (typical lateral size 50–300 nm, thickness 2–6 nm by AFM).
4. **Wash.** Dilute 1:3 with IPA, centrifuge 8000 rpm 10 min, redisperse in IPA. Repeat 2× to remove NMP.
5. **Concentrate.** Redisperse in IPA at **1 mg/mL**. Characterize by drop-casting on SiO_2/Si and AFM (target 3–8 layers).
6. **Yield:** ~5–15 % of starting mass as few-layer flakes; consistent with literature yields for bath sonication of BN in NMP.

2.2 Film deposition

7. **Substrate:** 5 × 5 mm^2 Si (300 nm SiO_2) or glass, cleaned (Sec. 1.10), UV-ozone 15 min.
8. **Drop-cast / spin.** For a continuous thin film: spin 200 µL of 1 mg/mL hBN/IPA at **2000 rpm, 60 s**, repeat 3× to build coverage. Anneal **200 °C, 30 min** in air to burn off organics and improve flake adhesion. - For **isolated single flakes** (preferred for single-defect addressing): dilute to 0.01 mg/mL, drop-cast 20 µL, air dry; flakes will be sparse, separable by > 5 µm.

2.3 V_B^- defect engineering

The negatively-charged boron vacancy, V_B^-, is a spin-1 (S=1) defect with zero-field splitting **D ≈ 3.46 GHz** and a characteristic **7-line ^{14}N hyperfine fingerprint** (coupling to 3 equivalent ^{14}N nuclei, A/h ≈ 47 MHz), photoluminescence centred near **850 nm** under 532 nm excitation [3][4][5]. Two creation routes:

Route A — Energetic irradiation (gold standard, recommended). V_B^- is reliably created by neutron irradiation [3] or focused-ion-beam (Xe/Ar/N) [5], followed by a vacuum activation anneal. This is the established, reproducible route; it requires access to an irradiation facility.

9. **Irradiation + activation anneal.** Irradiate the hBN film (neutron fluence ~10^{14} cm^{-2}, or FIB at ~suitable energy/dose), then anneal in **Ar or N_2** at **800–900 °C, 1–2 h** in the tube furnace to mobilise and activate the vacancies. This step *requires the tube furnace* — the 350 °C hotplate cannot reach activation temperature.

Route B — Chemical / thermal (maker-lab approximation, *unverified*). Wet-chemical creation of boron vacancies is *not an established route* — hBN is chemically inert to dilute HNO_3 and the

mechanism below is speculative. It is offered only as a low-cost approximation and should not be relied on to produce measurable VB− without validation against Route A.

10. **Wet-chemical attempt (unverified).** Soak hBN film in **0.1 M HNO_3** for **10 min**, rinse DI (intended to etch surface B atoms), then anneal in **Ar or N_2** at **500 °C, 1 h** in the tube furnace. Treat any ODMR signal as needing confirmation; do not assume B-vacancy formation.
11. **Carbon-related co-center creation.** Drop-cast **0.05 M citric acid in water** on the hBN, dry at 80 °C, anneal **600 °C, 30 min in Ar** (tube furnace). Carbon diffusion into B-vacancy sites creates carbon-related luminescent centres (broad emission ~580 nm) frequently reported alongside VB−.
12. **Charging to VB−.** Anneal under **N_2** (or forming gas 5 % H_2/N_2) at **400 °C, 30 min**, then quench to room temperature in < 30 s (lift off hotplate onto metal block). The negative charge state is stabilised by the Fermi level / local electrostatics and the rapid quench; hBN is stoichiometric BN, so do not assume "nitrogen donor" reservoirs analogous to P in Si.
13. **Defect density target:** ~0.1–0.3 /µm² (≈ 100–300 / 1000 µm²) — high enough to address ensembles for ODMR but low enough that single defects are resolvable in a confocal spot.

2.4 PMMA encapsulation

14. Spin **2 % PMMA 950K in anisole** at **4000 rpm, 45 s** → ~50 nm. Bake **180 °C, 2 min** on hotplate. PMMA serves as (a) oxygen/moisture barrier stabilizing VB− charge state, (b) index-matching layer (n ≈ 1.49) between hBN (n ≈ 2.1 out-of-plane effective ~1.7) and the TEOS sol-gel core (n ≈ 1.50), and (c) planarization for the subsequent waveguide.

Timing, Section 2: exfoliation 12 h + centrifugation/wash 3 h + film 1 h + defect engineering 4 h (anneal-dominated) + PMMA 1 h. **Total: ≈ 2 working days.**

Section 3 — On-Chip Fluorescence Photodetector Array (Graphene / MoS_2)

Goal: Place an **N-electrode graphene/MoS_2 photodetector array on-chip, aligned to the 8 × 8 qubit grid** — one detector electrode per addressed site, mirroring the emitter's N electrical contacts — and read it **directly by the FPGA**. No discrete photodiodes / Si APDs, and no external camera. The readout is entirely on-chip; coarse stacking only is needed because the emitter-pixel → qubit-site → detector-electrode mapping is recovered by **software pulse-capture calibration** (Section 5.6), so no micron-precision alignment jig is required.

3.1 CVD graphene transfer

1. **Source.** Commercial CVD graphene on Cu foil (Graphenea). Coat graphene face with **PMMA A4** (spin 2000 rpm, 60 s, ~200 nm), bake 180 °C, 2 min.

2. **Etch Cu.** Float on **0.5 M $FeCl_3$** (or ammonium persulfate 0.1 M) for 12 h; PMMA/graphene membrane releases. Rinse in 3 × DI water baths, 10 min each.
3. **Transfer.** Scoop onto target substrate (waveguide facet or pre-patterned Au electrode pair, 100 µm gap) using a bent Si wafer as a "fishing" scoop. Dry 24 h in air, then **180 °C, 1 h** to improve adhesion.
4. **Remove PMMA.** Soak in acetone 2 h, IPA rinse, N_2 blow dry. Some PMMA residue is acceptable.

3.2 MoS_2 transfer (optional, for higher-responsivity channel)

5. CVD monolayer MoS_2 on SiO_2/Si (commercial). Pattern Au contact pads first (Sec. 4.6), then transfer MoS_2 by **water-assisted delamination**: coat with PMMA, etch SiO_2 in **1 % HF** (50 s), scoop membrane onto target.
6. Anneal **200 °C, 2 h in Ar** to remove bubbles and residue.

3.3 Electrodes and channel definition

7. Photolithograph or shadow-mask Au/Cr (10/80 nm) source-drain electrodes with **100 µm channel** on the graphene/MoS_2.
8. Define channel by O_2 plasma (100 W, 30 s, 50 mTorr) through a photoresist window to isolate the graphene sheet into 4 detectors.

3.4 Operation

9. Bias drain–source **0.1–1 V**; responsivity typically **0.01–0.1 A/W (graphene, bare)**, **10–10^2 A/W (MoS_2 photogating with trap states)**; detectable down to nW optical powers with a lock-in. Use the same lock-in used for ODMR.

Section 4 — Waveguides, Layer Stacking, Bonding, Thermal Management, Contacts

4.1 TEOS sol-gel silica waveguide fabrication

> **Why not SU-8.** SU-8 is a novolac–epoxy photoresist with strong **autofluorescence** under visible excitation and C–H-overtone NIR absorption around 850 nm. For ODMR the VB− PL is at **~850 nm** with only **0.5–3 % contrast** on top of a 532 nm pump background — an autofluorescent waveguide would bury the signal. SU-8 also decomposes above ~300–400 °C, incompatible with the densification/post-anneal thermal budget. **TEOS-derived sol-gel SiO_2** is the correct material: fluorescence-free, UV–NIR transparent, and thermally stable. Core/clad index contrast is set by a silane dopant (MTEOS or Ti), not by an organic polymer.

1. **Core sol (n ≈ 1.50).** Mix **TEOS : EtOH : H_2O : HCl = 1 : 4 : 2 : 0.02 (mol)**; stir 2 h at 60 °C for hydrolysis/condensation. Add **15 mol % MTEOS** (methyltriethoxysilane) relative to TEOS — raises the refractive index and adds hydrophobicity. Age the sol 24 h at room temperature in a sealed vial; filter 0.22 µm PVDF.
2. **Cladding sol (n ≈ 1.45).** Same TEOS:EtOH:H_2O:HCl recipe, **no dopant**. Age 24 h, filter 0.22 µm.
3. **Core deposition + patterning.** On the PMMA-capped hBN substrate, spin core sol at **2000 rpm, 30 s** → ~0.5–1 µm; soft-bake **80 °C, 10 min**. Pattern waveguides **5–10 µm wide, 0.5–1 µm tall, 5–20 mm long, 1 cm radius bends** by **soft-lithography / UV embossing**: press a PDMS stamp (cast from a photolithographed or commercially-ruled master) into the wet sol, cure under 365 nm if a photo-initiator is added, release the stamp. (Shadow-mask + HF-vapor etch is an alternative — HF-vapor only in a fume hood with calcium-gluconate gel on hand.) Densify **300–400 °C, 1 h** on the hotplate or in the tube furnace → dense SiO_2; the index stabilises at ~1.50.
4. **Cladding.** Spin cladding sol at **1500 rpm, 30 s** → ~1 µm over the patterned core; bake 80 °C, 10 min; densify **300 °C, 30 min**. Δn ≈ 0.05 (core 1.50 / clad 1.45) — single or few-mode at both 532 nm and 850 nm. For a thicker cladding, repeat 2–3×.
5. **Propagation-loss target: < 0.5 dB/cm at 850 nm** (densified sol-gel silica, no organic fluorescence background). Sidewall roughness from the embossing stamp sets the practical loss; a reflow/rounding bake at 350 °C smooths edges.

4.2 Coupling the CQD emitter to the waveguide

5. Dice / cleave the CQD stack (Sec. 1) so the ASE gain-strip edge faces the sol-gel SiO_2 input facet. Mount the CQD chip and hBN+waveguide chip in the 3D-printed alignment jig. Use a 3-axis micrometer to bring the gain strip within < 5 µm of the sol-gel SiO_2 input facet.

6. UV-cure a small drop of NOA 61 optical adhesive (or cured PDMS as a "tack" layer) to lock the alignment. Butt-coupling efficiencies of **20–60 %** are realistic for end-fire coupling from an edge emitter into a 5–10 µm **TEOS sol-gel SiO_2** core when the gaps are < 2 µm and the NA is matched.

4.3 Layer stacking and bonding

7. **Stack order (bottom→top):** Si/SiO_2 carrier → Peltier TEC top plate → 5 × 5 mm hBN-on-SiO_2 chip → PMMA → TEOS sol-gel SiO_2 waveguides (core+clad) → CQD emitter chip (edge-coupled) → 2D-material photodetector at waveguide output.
8. Bond each layer with thin PDMS or cyanoacrylate at the corners only (avoid active area). Cure 80 °C, 1 h. Wire-bond or silver-paste the CQD anode/cathode and the photodetector source/drain to a PCB fan-out.

4.4 Peltier thermal management

9. **Module:** 2-stage TEC, e.g., TE Technology 9500/127/060 B (or generic TEC1-12706 single-stage for budget). Mount TEC cold side up, thermal-paste (Arctic Silver) to the Si carrier underside; TEC hot side to a finned aluminum heat sink with a small 40 mm fan.
10. **Control:** NTC 10 kΩ thermistor epoxied to the Si carrier next to the hBN chip. PID controller (e.g., Arduino + MAX6675/MCP9600 + MOSFET PWM) set to **295 K (22 °C)**, ±0.1 K stability. Operating range 288–300 K acceptable for room-temperature ODMR.
11. **Power:** TEC draws 1–3 A at 3–6 V from the bench supply; total chip dissipation < 5 W — compatible with a maker-lab power budget.

4.5 Contacts and encapsulation

12. Au wire bonds or silver-paste tracks to a 0.1" header for: ITO anode (×8 rows), Ag cathode (×8 cols), photodetector drains (×N, one electrode per addressed qubit site), gate (×1, optional electrolyte gate), thermistor (×2), TEC + and – (×2).
13. Encapsulate active area under a **1 mm PDMS slab** (oxygen/moisture barrier, optical clear). Leave waveguide output facet exposed for optional off-chip fiber coupling.

4.6 Coplanar waveguide (MW) for ODMR

14. Hand-etch a 50 Ω coplanar waveguide (G-S-G, signal 200 µm / gap 100 µm) on a single-sided FR4 or Rogers PCB. Mount the hBN chip face-down over the signal line (≤ 50 µm gap from chip backside to line) or flip-chip with anisotropic conductive paste. Microwave feedthrough to the on-board SMA launcher.

Section 5 — Characterization & Testing

5.1 Electroluminescence (CQD array)

1. Drive each pixel 0–15 V, log I–V and L–V. Confirm **turn-on 4–8 V** and broadband visible EL.
2. Spectra: couple edge emission to fiber + USB spectrometer. Expect broad Gaussian-like EL centered at the CQD PL peak (450–620 nm depending on synthesis).
3. **ASE check:** drive a single pixel with short pulses (1 ms, 10 % duty) while stepping current; watch for FWHM narrowing from ~80 nm to < 25 nm and a super-linear L–I knee. **Do not** interpret this as lasing — it is ASE.

5.2 ODMR of VB− in hBN

4. **Optical:** 532 nm CW laser (or the on-chip green CQD emitter), ~1 mW at the sample. Collect the VB− PL at **~850 nm** through a **650 nm red photographic long-pass filter** (passes the 850 nm NIR defect PL, blocks the 532 nm green pump) into the **on-chip graphene/MoS_2 detector array**. No Si APD / discrete photodiode — the readout is entirely on-chip.
5. **Microwave:** sweep 3.30–3.60 GHz from the function generator through the CPW. Modulate MW at 1 kHz for lock-in detection of the ODMR dip.
6. **Expected resonance:** the characteristic **7-line ^{14}N hyperfine pattern** centred at **D ≈ 3.46 GHz** (coupling to 3 equivalent ^{14}N, A/h ≈ 47 MHz, intensity ratios ≈ 1:3:6:7:6:3:1) [5]. ODMR contrast **0.5–3 %** (ensemble), linewidth **5–20 MHz** (broadened by strain and the ^{10}B/^{11}B nuclear bath). At low resolution the seven lines merge into a single dip near 3.46 GHz.
7. **Magnetic field sweep:** apply a static B_0 along the c-axis (a small permanent magnet on a linear stage); observe D shift following $D(B) = D_0 \pm \gamma_e \cdot B_0$ with $\gamma_e/2\pi \approx 28$ GHz/T. The doublet splits symmetrically — confirms the spin-1 (S=1) triplet of VB−.

5.3 Spin coherence (T_2)

8. **Pulsed ODMR** (requires pulse-capable source or fast PIN switch): run Hahn echo ($\pi/2 - \tau - \pi - \tau -$ echo) at the resonance. Vary τ from 0.1–20 µs. Fit echo amplitude vs τ to $\exp[-(2\tau/T_2)^n]$ with n ≈ 1–2.
9. **Expected T_2: 0.5–5 µs at room temperature** for ensembles in exfoliated hBN, limited by ^{10}B/^{11}B nuclear spins and paramagnetic impurities. T_1 typically 10–100 µs.

5.4 Coupling efficiency (waveguide → detector)

10. Inject calibrated light (e.g., 532 nm, 1 µW) into the sol-gel SiO_2 input facet with a tapered fiber; measure power at the output facet with a power meter and the on-chip graphene detector photocurrent.

11. **Coupling efficiency target: 20–60 %** (input) and propagation loss < 1 dB/cm; on-chip detector responsivity measured from the known input power.

5.5 End-to-end functional test

12. Drive CQD pixel electrically → green EL pumps the hBN V_B^- ensemble → collect the ~850 nm defect PL (through the 650 nm long-pass filter) at the on-chip graphene/MoS_2 detector → its photocurrent is modulated by MW at D ≈ 3.46 GHz → the FPGA / lock-in reads the ODMR dip. A successful end-to-end ODMR with the on-chip CQD pump (no external laser) is the chip's defining demonstration.

5.6 Emitter ↔ qubit ↔ detector calibration (software pulse-capture)

13. **No manual pixel-perfect alignment is required.** With the stack coarse-assembled (layer-to-layer registration ~100 µm is enough), run an automated calibration sweep from the FPGA: for each emitter pixel (i,j) fire a short green pump pulse, then sample every graphene/MoS_2 detector electrode and record which one(s) show a modulated ~850 nm response under MW at D ≈ 3.46 GHz. The sweep builds a lookup table mapping **emitter pixel → qubit site → detector electrode**. Subsequent operation addresses qubits through this table, so the chip tolerates coarse mechanical stacking and needs **no camera and no micron alignment jig for readout**. (A camera on the separate FPGA controller board, if present, is used only for initial visual inspection of the assembled chip — never for qubit readout.)

Section 6 — Safety

HAZARD	CONTROL
HNO_3, HCl, HF (1 %)	Fume hood, nitrile + butyl gloves, face shield; HF → calcium gluconate gel on hand
NMP, anisole, acetone	Flammable; fume hood; no ignition sources
Hydrothermal autoclave 200 °C, pressurized	Never open hot; let cool < 60 °C; rated PTFE liner; eye protection
UV 365 nm exposure	UV-blocking glasses; skin cover
Thermal evaporator	Interlock; cool before opening; Au/Cr evaporation produces vapor
532 nm laser 50 mW	Class 3B; laser goggles OD > 4 at 532 nm; beam stops
MW up to 6 GHz, < +20 dBm	Low risk; avoid direct contact with antenna; keep > 30 cm
Peltier / hot surfaces	Heat-resistant gloves; allow hotplate cool-down
Nanomaterials (CQDs, hBN flakes)	Treat as nuisance dust; N95; wet-clean surfaces; dispose as solid nano-waste
Cu etchant $FeCl_3$	Corrosive; gloves; stain hazard

General: Maintain an SDS binder for every chemical. Eye wash station and emergency shower accessible. No food/drink. Solvent and acid waste in separate labeled carboys.

7. Expected Performance Table

PARAMETER	TARGET / EXPECTED	BASIS
CQD PL peak (solution)	450–620 nm (tunable)	Sec. 1.1, lit.
CQD quantum yield (solution)	40–80 %	Sec. 1.1
CQD EL turn-on voltage	4–8 V	Sec. 1.7, 5.1
CQD EL radiance	≥ 10 cd/m²	Sec. 5.1
ASE threshold (edge)	detectable, FWHM < 25 nm above threshold	Sec. 5.1
CQD lasing	**NOT claimed**	see note
hBN flake thickness	2–4 nm (3–8 layers)	Sec. 2.1
VB− zero-field splitting D	3.46 GHz	Sec. 2.3, 5.2
Defect density (VB−)	0.1–0.3 /µm²	Sec. 2.3
ODMR contrast (ensemble)	0.5–3 %	Sec. 5.2
ODMR linewidth	5–20 MHz	Sec. 5.2
Spin coherence T_2 (RT)	0.5–5 µs	Sec. 5.3
Spin relaxation T_1 (RT)	10–100 µs	Sec. 5.3
Waveguide coupling	20–60 %	Sec. 4.2, 5.4
Sol-gel SiO_2 propagation loss	< 0.5 dB/cm @ 850 nm (fluorescence-free)	Sec. 4.1
Graphene photodetector responsivity	0.01–0.1 A/W	Sec. 3.4
MoS_2 photodetector responsivity	10–10^2 A/W (photogating)	Sec. 3.4
Operating temperature	288–300 K (set 295 K)	Sec. 4.4
TEC power	< 5 W	Sec. 4.4
Total chip footprint	25 × 25 mm²	Sec. 2.4

8. Validation Against Known Studies

For each major claim, the supporting peer-reviewed literature is listed. Where the claim is at the edge of what has been reported, the wording is hedged. References are numbered and listed in the References section below.

11.1 CQD hydrothermal synthesis — yield, wavelength, QY

- **Citric acid + urea/ethylenediamine hydrothermal route** is the canonical CQD synthesis, producing blue-to-green PL with QY 40–80 %. Consistent with results reported in **Baker & Baker, *Angew. Chem. Int. Ed.* 2010** [1] (landmark review of luminescent carbon nanodots) and the large CQD literature summarized in **Lim, Shen & Gao, *Chem. Soc. Rev.* 2015** [2]. The 200 °C, 6 h hydrothermal parameters used here are within the standard ranges reported.
- **Wavelength tunability** via nitrogen dopant ratio and hydrothermal temperature is a broadly reported property of CQDs, consistent with the reviews above and with numerous primary reports.

11.2 CQD electroluminescence and ASE

- **Electroluminescence from CQD-polymer composites** under modest DC bias (turn-on ~5–10 V) has been reported by several groups; consistent with the CQD LED literature summarized in **Lim et al. 2015** [2] and subsequent CQD-LED reviews.
- **ASE / amplified spontaneous emission from CQDs under optical pumping** is well documented, e.g., white-light ASE from carbon nanodots under laser excitation **Zhang et al., *Opt. Mater. Express* 2012** [29], and near-unity-QY CQDs for electroluminescent LEDs **Yuan et al., *Angew. Chem. Int. Ed.* 2023** [30].
- **Electrically-pumped CQD *lasing* is NOT claimed.** True electrically-pumped lasing from CQDs has been pursued but remains difficult and controversial; we do not cite any paper as having demonstrated it unambiguously. We deliberately limit the claim to **EL and ASE**. ASE is a necessary but not sufficient precondition for lasing and does not require a coherent cavity mode.

11.3 Liquid exfoliation of hBN

- **NMP-based bath sonication** to produce few-layer hBN is the standard liquid-phase exfoliation route, established in **Hernandez et al., *Nat. Nanotechnol.* 2008** [3] (one-step mass production of few-layer graphene by liquid exfoliation; the method generalizes to BN, MoS_2, etc.) and in **Coleman et al., *Science* 2011** [4] ("Two-Dimensional Nanosheets Produced by Liquid Exfoliation of Layered Materials"), which explicitly reports few-layer BN in NMP with the surface-tension-matching criterion we use. The yields (~5–15 % few-layer) and flake sizes (50–300 nm) targeted here are consistent with those reports.

11.4 VB− defect, D ≈ 3.46 GHz, room-temperature spin

- **Discovery and room-temperature optically-detected magnetic resonance of the negatively-charged boron vacancy VB− in hBN**, with zero-field splitting **D ≈ 3.46 GHz**, optical initialization and readout: **Gottscholl et al., *Nature Materials* 19, 540–545 (2020)** [5]. This is the landmark paper for the VB− qubit and directly supports Section 2.3 and the ODMR target in Section 5.2.
- **Theoretical identification of the VB− structure and spin Hamiltonian**, including D ≈ 3.46 GHz: **Ivády et al., *npj Computational Materials* 6, 41 (2020)** [6] (ab initio theory of the negatively charged boron vacancy qubit in hBN). Supports the spin-1 triplet assignment and the magnetic-field splitting used in Section 5.2.
- **Room-temperature spin coherence (T_2 ~ µs)** of VB− ensembles: consistent with values reported in **Gottscholl et al. 2020** [5] and follow-up work by **Kianinia et al.** [7] and **Chejanovsky et al.** [8] on hBN spin defects and single-photon emitters. µs-scale T_2 at room temperature is the accepted performance envelope for as-grown/implanted VB− ensembles; isotopic engineering (^{10}B / ^{12}C purification) extends it but is outside maker-lab scope.
- **Defect density ~0.3 /µm²** is within the range routinely reported for irradiated or annealed hBN samples used for ensemble ODMR; the chemical route in Section 2.3 is a maker-lab approximation of ion-implantation results in [5–7].
- **hBN as a quantum-emitter host**: single-photon emission from hBN monolayers **Tran et al., *Nat. Nanotechnol.* 2016** [26], hBN photonics review **Caldwell et al., *Nat. Rev. Mater.* 2019** [27], and a quantum-coherent spin in hBN at ambient conditions **Stern et al., *Nat. Mater.* 2024** [28] (a carbon-related defect, distinct from VB−).

11.5 Waveguide coupling (20–60 %)

- **Butt/end-fire coupling of edge emitters into micrometer-scale silica waveguides** with 20–60 % efficiency is standard in integrated photonics and consistent with values reported in sol-gel silica and polymer-waveguide quantum-photonic work; see polymer integrated optics **Bêche et al., *Microelectron. J.* 2006** [31] and soft-lithography patterning **Xia & Whitesides 1998** [32], **Qin et al., *Nat. Protoc.* 2010** [33]. The 5–10 µm **TEOS sol-gel SiO_2** core and < 2 µm gap specified in Section 4.2 are within the regime where end-fire efficiency is dominated by mode-overlap and routinely reaches tens of percent. TEOS sol-gel silica is preferred over SU-8 here because its **negligible autofluorescence** under 532 nm excitation is essential for reading the weak (~0.5–3 % contrast) V_B^- ODMR PL at 850 nm without waveguide-background burial.

11.6 On-chip 2D-material photodetector responsivity

- **Graphene photodetectors**: **Xia et al., *Nat. Nanotechnol.* 2009** [9] (ultrafast graphene photodetector) and subsequent reviews establish bare-graphene responsivities of **0.01–0.1 A/W**, matching our target.
- **MoS_2 and hybrid 2D photodetectors with high responsivity (10–10^2 A/W) via photogating**: **Konstantatos et al., *Nat. Nanotechnol.* 2012** [10] (hybrid graphene–quantum-dot phototransistor

with ultrahigh gain), ultrasensitive monolayer-MoS_2 photodetectors **Lopez-Sanchez et al., *Nat. Nanotechnol.* 2013** [24], and vertical graphene heterostructure transistors **Britnell et al., *Science* 2012** [25] establish the photogating regime used in Section 3.4.

- **Integration of 2D photodetectors on waveguides** for on-chip readout is consistent with 2D-material photodetection reviews (e.g., F. Xia et al., *Nat. Photonics* reviews).

11.7 Peltier / TEC cooling for stable room-temperature operation

- Thermoelectric (Peltier) cooling to stabilize solid-state photonics and single-photon detectors to ±0.1 K is standard engineering; no specific landmark citation is required. Generic thermoelectric-cooler references and vendor application notes suffice. The ±0.1 K stability in Section 4.4 is well within commercial TEC + PID capability.

11.8 Related defect platforms (NV-diamond, SiC V_Si) and decoupling

The chip is VB−-centric, but the same control stack drives other room-temperature solid-state spin defects. These references support that generality and the calibration/decoupling toolbox:

- **NV centre in diamond** (D ≈ 2.87 GHz): the canonical room-temperature optically-addressable spin, reviewed by **Doherty et al., *Phys. Rep.* 2013** [11]; nanoscale magnetometry **Balasubramanian et al., *Nature* 2008** [12]; single-spin coherent oscillations **Jelezko et al., *PRL* 2004** [13]; nanoscale magnetic sensing **Maze et al., *Nature* 2008** [14]; and near-second coherence under decoupling **Bar-Gill et al., *Nat. Commun.* 2013** [23].
- **Silicon vacancy V_Si in 4H-SiC** (S = 3/2, $2|D| \approx 70$ MHz, ZPL ≈ 916 nm): room-temperature coherent control **Widmann et al., *Nat. Mater.* 2015** [16] and **Kraus et al., *Nat. Phys.* 2014** [15]; the V_Si as a quantum system **Baranov et al., *PRB* 2011** [19]; single V1 dynamics **Morioka et al., *Phys. Rev. Applied* 2022** [20]. Note: **Christle et al., *Nat. Mater.* 2015** [18] and **Christle et al., *PRX* 2017** [17] report millisecond coherence for the **SiC divacancy** (V_SiV_C, S = 1), *not* the silicon vacancy — they are cited here for the divacancy, a distinct defect.
- **Dynamical decoupling** used to extend T_2: **de Lange et al., *Science* 2010** [21] and **Ryan et al., *PRL* 2010** [22].
- **Microwave delivery**: a low-cost resonator approach for ODMR excitation **Bülau et al., *Metrology* 2025** [34].

End of protocol. This document is a maker-lab demonstrator specification. It is not a medical, military, or safety-critical device specification. Where primary literature is cited, the cited paper should be consulted before relying on a specific numerical parameter in a downstream application.

References

[1] S. N. Baker and G. A. Baker, "Luminescent carbon nanodots: emergent nanolights," Angewandte Chemie International Edition 49, 6726-6744 (2010). doi:10.1002/anie.200906623

[2] S. Y. Lim, W. Shen, and Z. Gao, "Carbon quantum dots and their applications," Chemical Society Reviews 44, 362-381 (2015). doi:10.1039/C4CS00269E

[3] Y. Hernandez, V. Nicolosi, M. Lotya, F. M. Blighe, et al., "High-yield production of graphene by liquid-phase exfoliation of graphite," Nature Nanotechnology 3, 563-568 (2008). doi:10.1038/nnano.2008.215

[4] J. N. Coleman, M. Lotya, A. O'Neill, et al., "Two-dimensional nanosheets produced by liquid exfoliation of layered materials," Science 331, 568-571 (2011). doi:10.1126/science.1194975

[5] A. Gottscholl, M. Kianinia, V. Soltamov, S. Orlinskii, G. Mamin, C. Bradac, C. Kasper, K. Krambrock, A. Sperlich, M. Toth, I. Aharonovich, and V. Dyakonov, "Initialization and read-out of intrinsic spin defects in a van der Waals crystal at room temperature," Nature Materials 19, 540-545 (2020). doi:10.1038/s41563-020-0619-6

[6] V. Ivády, G. Barcza, G. Thiering, S. Li, H. Hamdi, J.-P. Chou, Ö. Legeza, and A. Gali, "Ab initio theory of the negatively charged boron vacancy qubit in hexagonal boron nitride," npj Computational Materials 6, 41 (2020). doi:10.1038/s41524-020-0305-x

[7] M. Kianinia, S. White, J. E. Fröch, C. Bradac, and I. Aharonovich, "Generation of spin defects in hexagonal boron nitride," ACS Photonics 7, 2147-2152 (2020). doi:10.1021/acsphotonics.0c00614

[8] N. Chejanovsky, A. Mukherjee, J. Geng, Y.-C. Chen, Y. Kim, A. Denisenko, A. Finkler, T. Taniguchi, K. Watanabe, D. B. R. Dasari, P. Auburger, A. Gali, J. H. Smet, and J. Wrachtrup, "Single-spin resonance in a van der Waals embedded paramagnetic defect," Nature Materials 20, 1079-1084 (2021). doi:10.1038/s41563-021-00979-4

[9] F. Xia, T. Mueller, Y.-M. Lin, A. Valdes-Garcia, and P. Avouris, "Ultrafast graphene photodetector," Nature Nanotechnology 4, 839-843 (2009). doi:10.1038/nnano.2009.292

[10] G. Konstantatos, M. Badioli, L. Gaudreau, J. Osmond, M. Bernechea, F. P. Garcia de Arquer, F. Gatti, and F. H. L. Koppens, "Hybrid graphene-quantum dot phototransistors with ultrahigh gain," Nature Nanotechnology 7, 363-368 (2012). doi:10.1038/nnano.2012.60

[11] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, "The nitrogen-vacancy colour centre in diamond," Physics Reports 528, 1-45 (2013). doi:10.1016/j.physrep.2013.02.001

[12] G. Balasubramanian, I. Y. Chan, R. Kolesov, et al., "Nanoscale imaging magnetometry with diamond spins under ambient conditions," Nature 455, 648-651 (2008). doi:10.1038/nature07278

[13] F. Jelezko, T. Gaebel, I. Popa, A. Gruber, and J. Wrachtrup, "Observation of coherent oscillations in a single electron spin," Physical Review Letters 92, 076401 (2004). doi:10.1103/PhysRevLett.92.076401

[14] J. R. Maze, P. L. Stanwix, J. S. Hodges, et al., "Nanoscale magnetic sensing with an individual electronic spin in diamond," Nature 455, 644-647 (2008). doi:10.1038/nature07279

[15] H. Kraus, V. A. Soltamov, D. Riedel, S. Väth, F. Fuchs, A. Sperlich, P. G. Baranov, V. Dyakonov, and G. V. Astakhov, "Room-temperature quantum microwave emitters based on spin defects in silicon carbide," Nature Physics 10, 157-162 (2014). doi:10.1038/nphys2826

[16] M. Widmann, S.-Y. Lee, T. Rendler, et al., "Coherent control of single spins in silicon carbide at room temperature," Nature Materials 14, 164-168 (2015). doi:10.1038/nmat4145

[17] D. J. Christle, P. V. Klimov, C. F. de las Casas, et al., "Isolated spin qubits in SiC with a high-fidelity infrared spin-to-photon interface," Physical Review X 7, 021046 (2017). doi:10.1103/PhysRevX.7.021046

[18] D. J. Christle, A. L. Falk, P. Andrich, P. V. Klimov, J. U. Hassan, N. T. Son, E. Janzén, T. Ohshima, and D. D. Awschalom, "Isolated electron spins in silicon carbide with millisecond coherence times," Nature Materials 14, 160-163 (2015). doi:10.1038/nmat4144

[19] P. G. Baranov, A. P. Bundakova, A. A. Soltamova, S. B. Orlinskii, I. V. Borovykh, R. Zondervan, R. Verberk, and J. Schmidt, "Silicon vacancy in SiC as a promising quantum system for single-defect and single-photon spectroscopy," Physical Review B 83, 125203 (2011). doi:10.1103/PhysRevB.83.125203

[20] N. Morioka, D. Liu, Ö. O. Soykal, I. Gediz, C. Babin, R. Stöhr, T. Ohshima, N. T. Son, J. Ul-Hassan, F. Kaiser, and J. Wrachtrup, "Spin-optical dynamics and quantum efficiency of a single V1 center in silicon carbide," Physical Review Applied 17, 054005 (2022). doi:10.1103/PhysRevApplied.17.054005

[21] G. de Lange, Z. H. Wang, D. Ristè, V. V. Dobrovitski, and R. Hanson, "Universal dynamical decoupling of a single solid-state spin from a spin bath," Science 330, 60-63 (2010). doi:10.1126/science.1192739

[22] C. A. Ryan, J. S. Hodges, and D. G. Cory, "Robust decoupling techniques to extend quantum coherence in diamond," Physical Review Letters 105, 200402 (2010). doi:10.1103/PhysRevLett.105.200402

[23] N. Bar-Gill, L. M. Pham, A. Jarmola, D. Budker, and R. L. Walsworth, "Solid-state electronic spin coherence time approaching one second," Nature Communications 4, 1743 (2013). doi:10.1038/ncomms2771

[24] O. Lopez-Sanchez, D. Lembke, M. Kayci, A. Radenovic, and A. Kis, "Ultrasensitive photodetectors based on monolayer MoS2," Nature Nanotechnology 8, 497-501 (2013). doi:10.1038/nnano.2013.100

[25] L. Britnell, R. V. Gorbachev, R. Jalil, et al., "Field-effect tunneling transistor based on vertical graphene heterostructures," Science 335, 947-950 (2012). doi:10.1126/science.1218461

[26] T. T. Tran, K. Bray, M. J. Ford, M. Toth, and I. Aharonovich, "Quantum emission from hexagonal boron nitride monolayers," Nature Nanotechnology 11, 37-41 (2016). doi:10.1038/nnano.2015.242

[27] J. D. Caldwell, I. Aharonovich, G. Cassabois, J. H. Edgar, B. Gil, and D. N. Basov, "Photonics with hexagonal boron nitride," Nature Reviews Materials 4, 552-567 (2019). doi:10.1038/s41578-019-0124-1

[28] H. L. Stern, C. M. Gilardoni, Q. Gu, et al., "A quantum coherent spin in hexagonal boron nitride at ambient conditions," Nature Materials 23, 1379-1385 (2024). doi:10.1038/s41563-024-01887-z

[29] W. F. Zhang, L. B. Tang, S. F. Yu, and S. P. Lau, "Observation of white-light amplified spontaneous emission from carbon nanodots under laser excitation," Optical Materials Express 2, 490-495 (2012). doi:10.1364/OME.2.000490

[30] T. Yuan, F. Yuan, L. Sui, Y. Zhang, Y. Li, X. Li, Z. Tan, and L. Fan, "Carbon quantum dots with near-unity quantum yield bandgap emission for electroluminescent light-emitting diodes," Angewandte Chemie International Edition 62, e202218568 (2023). doi:10.1002/anie.202218568

[31] B. Bêche, N. Pelletier, E. Gaviot, et al., "Conception of optical integrated circuits on polymers," Microelectronics Journal 37, 421-427 (2006). doi:10.1016/j.mejo.2005.07.003

[32] Y. Xia and G. M. Whitesides, "Soft lithography," Annual Review of Materials Research 28, 153-184 (1998). doi:10.1146/annurev.matsci.28.1.153

[33] D. Qin, Y. Xia, and G. M. Whitesides, "Soft lithography for micro- and nanoscale patterning," Nature Protocols 5, 491-502 (2010). doi:10.1038/nprot.2009.234

[34] A. R. Bülau et al., "A highly efficient, low-cost microwave resonator for exciting a diamond sample from a miniaturized quantum magnetometer," Metrology 5, 69 (2025). doi:10.3390/metrology5040069
